PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: SettleOps AI

Version: 1.0

1. Project Overview

- **Problem Statement:** Insurance claim review is still heavily manual, requiring claims officers to read and reconcile multiple documents (police reports, adjuster reports, policy terms, repair quotations, and photos). This process:
    - Takes 30–60 minutes per claim
    - Causes backlogs and delays
    - Leads to inconsistent decisions
    - Increases risk of missed fraud signals
- **Target Domain:** An AI-powered decision support system for motor insurance claims that integrates document intelligence, multi-source reasoning, and fraud and liability assessment.
- **Proposed Solution Summary:** SettleOps AI is a multi-agent AI automation workflow platform that analyzes claim evidence, orchestrates intelligent agent reasoning, and generates structured, transparent decisions in under 90 seconds with human-in-the-loop control.

2. Background & Business Objective

- **Background of the Problem:** For many years, motor insurance claim review has followed a manual, fragmented workflow. A claims officer must read multiple sources —

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police reports, adjuster reports, policy documents, repair quotations, and crash photos, and mentally reconcile them to make a decision. This process requires extensive manual effort, constant context switching between systems, and heavy reliance on individual experience. As a result, decision-making is slow, inconsistent, and prone to missed fraud signals, especially under high claim volumes.

- **Importance of Solving This Issue:** By automating claim analysis through a multi-agent AI workflow, the time required to produce a decision can be reduced from 30–60 minutes to under 90 seconds. This allows insurers to process claims at scale, improve consistency, and detect fraud more effectively through cross-document reasoning. It also reduces cognitive load on claims officers, enabling them to focus on supervising decisions rather than manually constructing them, ultimately accelerating claim settlements and reducing operational costs.
- **Strategic Fit / Impact:** SettleOps AI aligns with the emerging paradigm of AI agent-based automation workflows, where multiple specialized agents collaborate to solve complex, real-world tasks. It complements existing insurance ecosystems (e.g., claims workflow platforms) by adding a dedicated AI reasoning layer rather than replacing current systems. This positions the product as a scalable intelligence layer for the insurance industry, enabling faster, more transparent, and auditable decision-making while supporting future expansion into fully automated claim processing.

3. Product Purpose

3.1. Main Goal of the System: To provide an AI agent-based automation workflow that helps insurance claims officers analyze claim evidence, coordinate reasoning across multiple specialized agents, and generate a structured claim decision quickly with human supervision.

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3.2. Intended Users (Target Audience):

- **Claims Officers**: The primary users who review claim evidence, inspect AI reasoning, and make the final decision.
- **Claims Team Leads / Supervisors**: Users who monitor escalated cases, review decision quality, and use audit trails for governance.
- **Head of Claims Operations**: Business buyer who wants faster processing, better consistency, and higher throughput.
- **Fleet Operations Managers / Drivers**: Secondary stakeholders who benefit from faster claim settlement and reduced vehicle downtime.

4. System Functionalities

4.1. Description: The system operates as an AI agent-based automation workflow engine for insurance claim decision-making. It ingests multi-source claim evidence (documents, images, reports), coordinates multiple specialized AI agents to analyze different aspects of the case in parallel, and synthesizes their outputs into a structured, auditable decision. The system emphasizes transparent reasoning, real-time workflow visibility, and human-in-the-loop control, allowing claims officers to supervise, challenge, and finalize AI-generated decisions.

4.2. Key Functionalities:

- **Multi-Agent Claim Analysis:** Coordinates 9–13 specialized AI agents to analyze claim evidence in parallel and produce a unified decision:

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- **Intake Specialist:** Categorizes documents (Police reports, Policy books, Adjuster logs) into the correct slots.
- **Policy Specialist:** Extracts coverage type, maximum payouts, excess MYR, and depreciation rates.
- **Liability Adjuster:** Extracts incident time, location, and the narrative sequence from police reports.
- **Vision AI Agent:** A specialized "Visual Forensic Analyst" that determines the Point of Impact (POI) and damage severity from crash photos.
- **Damage Assessor:** Audits workshop repair quotes for part cost consistency and labor necessity.
- **Fraud Investigator:** Performs cross-document analysis to identify suspicious patterns or narrative contradictions.
- **Payout Strategist:** A deterministic agent that reconciles verified damage, liability rates, and policy terms to calculate the final MYR payout.
- **Senior Auditor:** The high-level supervisor that performs the final cross-consistency check and flags conflicts for reruns.
- **Feedback Alignment Agent:** Translates freeform human chat from the "AI Strategy Chat" into structured instructions for other agents.
- **Validation Gate:** A deterministic agent that ensures "Intake Health" before allowing the analysis to proceed.
- **Report Drafter Agent:** Finalizes the decision artifact PDF with the aggregated reasoning.

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- **Workflow Orchestrator:** The central LangGraph controller that manages parallel execution and the surgical feedback loops.
- **SSE Streaming Agent:** Manages the real-time "Blackboard" updates and agent status tickers to keep the UI alive.

- **Transparent Reasoning & Workflow Visualization:** Displays AI reasoning steps, including status (active, completed, challenged, escalated), ensuring decisions are explainable and auditable.
- **Human-in-the-Loop Control:** The system allows claims officers to remain the final decision-makers by providing clear action options to approve decisions, decline them with proper justification, or challenge specific components of the AI's reasoning. When a challenge is triggered, the system initiates a targeted re-analysis of the affected reasoning area, ensuring that decisions are not only fast but also accurate and reviewable.
- **AI Strategy Chat (Conversational Intervention):** The AI Strategy Chat enables claims officers to interact directly with the system in a conversational manner. Officers can ask for explanations behind specific decisions, request recalculations based on new assumptions, or provide feedback to refine the AI's reasoning. This interaction enhances transparency and allows users to better understand and guide the decision-making process.
- **Fraud Detection & Escalation:** Identifies inconsistencies across documents and flags potential fraud cases, triggering escalation when risk thresholds are exceeded.
- **Decision Output & Audit Trail Generation:** The system automatically generates a structured Claim Decision PDF along with a detailed audit trail. These outputs include key elements such as claim summaries, coverage analysis,

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liability assessment, fraud indicators, payout breakdown, and decision history. The audit trail ensures that every step of the AI reasoning and user interaction is recorded, supporting compliance, review, and operational accountability.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: SettleOps AI utilizes Google Gemini 2.5 Flash as its foundational intelligence engine. Our benchmarking against industry leaders revealed that while several models offer high reasoning capability, Gemini 2.5 Flash provides the optimal balance of speed, multimodal vision, and context length required for real-time insurance automation.

Model	Reasoning Depth	Inference Speed	Context Window	Vision Accuracy	Cost Efficiency	SettleOps Fit
Gemini 2.5 Flash	High	Ultra High	1M+ Tokens	Excellent	High	Primary Engine
OpenAI GPT-4o	Very High	Fast	128k	Good	Moderate	Secondary/ Fallback
Claude 3.5 Sonnet	Very High	Moderate	200k	Moderate	Low (Costly)	Over-engineered
GLM-4	Moderate	Fast	128k	Basic	Moderate	Regional alternative
Gemini 1.5 Pro	Ultra High	Moderate	2M+ Tokens	Excellent	Moderate	Auditor Tasks

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Why Gemini 2.5 Flash?

- **Sub-90s Decisions:** Its high token-per-second throughput is critical for our 90-second generation target.
- **Multimodal Vision AI:** Native image support allows the Vision AI Agent to analyze crash photos directly for damage corroboration without external OCR dependencies.
- **Infinite Context:** Handling multiple PDFs (police reports, adjuster reports, and 100-page policy booklets) simultaneously requires a massive context window which Gemini provides.

Development vs. Production Strategy During the development and prototyping phase, the team utilized Gemini 1.5 Pro and Claude 3.5 Sonnet for their extreme reasoning depth. This allowed us to validate the complex multi-agent architecture and "stress-test" the liability logic across contradictory evidence sets. For a live Production environment, however, we prioritize Gemini 2.5 Flash. The shift to 2.5 Flash is driven by the need for commercial scalability; it delivers the rapid inference speeds required for a responsive user experience and a significantly more sustainable cost-per-claim ratio, which is essential for handling the high volume of claims in the Malaysian insurance market.

4.3.2. Prompting Strategy: The system adopts a multi-step agentic prompting strategy, where complex claim evaluation is decomposed into smaller, specialized tasks handled by dedicated AI agents such as liability, fraud detection, coverage analysis, and payout calculation. Each agent operates with a focused prompt tailored to its responsibility, ensuring depth and precision in its analysis. The outputs from these agents are then passed to a higher-level Auditor Agent, which validates, cross-checks, and synthesizes the results into a coherent final recommendation. This approach is particularly suitable for insurance claims because the problem inherently involves multiple interdependent factors and sources of truth. By breaking the workflow into structured sub-tasks, the system improves accuracy, enhances traceability of

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reasoning, and enables parallel processing, ultimately delivering faster and more reliable decision outcomes.

4.3.3. Context & Input Handling: The system is designed to handle large volumes of unstructured and heterogeneous inputs, including lengthy documents, images, and multi-source evidence. It achieves this through a combination of document parsing, intelligent chunking, and context aggregation. Long documents are split into manageable segments to fit within model limits, while key information is extracted and structured before being passed into the reasoning pipeline. The system then aggregate relevant context across all evidence sources, allowing agents to reason holistically rather than in isolation. For inputs that exceed model capacity, the system processes them sequentially and retains important insights in a structured intermediate memory, ensuring continuity across reasoning steps. Non-critical or redundant information may be deprioritized to optimize performance, but essential details are preserved to maintain decision accuracy.

4.3.4. Fallback & Failure Behavior: To ensure reliability and mitigate risks such as hallucinations or incorrect reasoning, the system incorporates multiple layers of validation and control. The Auditor Agent plays a central role by detecting inconsistencies, weak justifications, or conflicting evidence before any recommendation is finalized. Additionally, a built-in challenge mechanism allows claims officers to question specific parts of the AI's reasoning, triggering targeted re-evaluation. When uncertainty cannot be resolved within defined limits, the system follows an escalation path, presenting the case to the user with clear indications of unresolved issues rather than forcing a potentially incorrect decision. Furthermore, the system enforces controlled output behavior, ensuring that low-confidence results are not presented as definitive conclusions. Overall, the design prioritizes safety, transparency, and human oversight, ensuring that all outputs remain explainable, reviewable, and trustworthy.

5. User Stories & Use Cases

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- **User Stories:**
    - As a claims officer, I want to upload all claim documents and photos in one place so that I can start the review process quickly without switching between systems.
    - As a claims officer, I want the AI to analyze all evidence and generate a structured decision so that I do not need to manually read and reconcile every document.
    - As a claims officer, I want to see how the AI reaches its decision so that I can trust and verify the reasoning before approving.
    - As a claims officer, I want to challenge specific parts of the AI's reasoning so that I can correct or refine the outcome when needed.
    - As a claims officer, I want to interact with the AI through chat so that I can ask questions, request explanations, and adjust assumptions easily.
    - As a claims team lead, I want to review escalated cases and audit trails so that I can ensure decision quality and compliance.
    - As a head of claims operations, I want faster and more consistent claim decisions so that overall operational efficiency and throughput improve.
    - As a fleet operations manager, I want claims to be processed faster so that vehicle downtime is minimized.
- **Use Cases (Main Interactions):**

  Claim Submission & Case Initiation Flow: The user uploads claim documents (police report, adjuster report, policy, photos) through the system interface. The system validates the inputs and creates a new claim case, which then appears in the claims queue and triggers the AI review workflow.

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AI Multi-Agent Analysis Flow: Once a case is initiated, the system activates multiple specialized AI agents to process the evidence in parallel. Each agent analyzes a specific aspect (e.g., liability, fraud, coverage), and their outputs are continuously updated and displayed in the workflow panel. The Auditor Agent then synthesizes these outputs into a structured recommendation.

Decision Review & Interaction Flow: The claims officer opens a case in the three-pane dashboard, where they can view all inputs, monitor the live reasoning workflow, and inspect the structured decision blackboard. The officer can interact with the AI through the strategy chat to ask for explanations or request recalculations, ensuring full transparency and control.

Challenge & Re-analysis Flow: If the officer disagrees with any part of the AI's reasoning, they can challenge a specific component (e.g., liability percentage or payout). The system then triggers a targeted re-analysis of that component, updates the results, and reflects the changes in the dashboard.

Final Decision & Output Generation Flow: After reviewing the AI recommendation, the claims officer can approve or decline the claim. Upon approval, the system generates a Claim Decision PDF along with a structured audit trail documenting the reasoning process, decisions made, and any challenges raised. In a real production environment, the system can also support electronic digital signatures compliant with Malaysian legal frameworks (e.g., Digital Signature Act), allowing officers to formally sign and validate decisions in a legally recognized manner before final submission and record keeping.

- **Use Case Diagram:**

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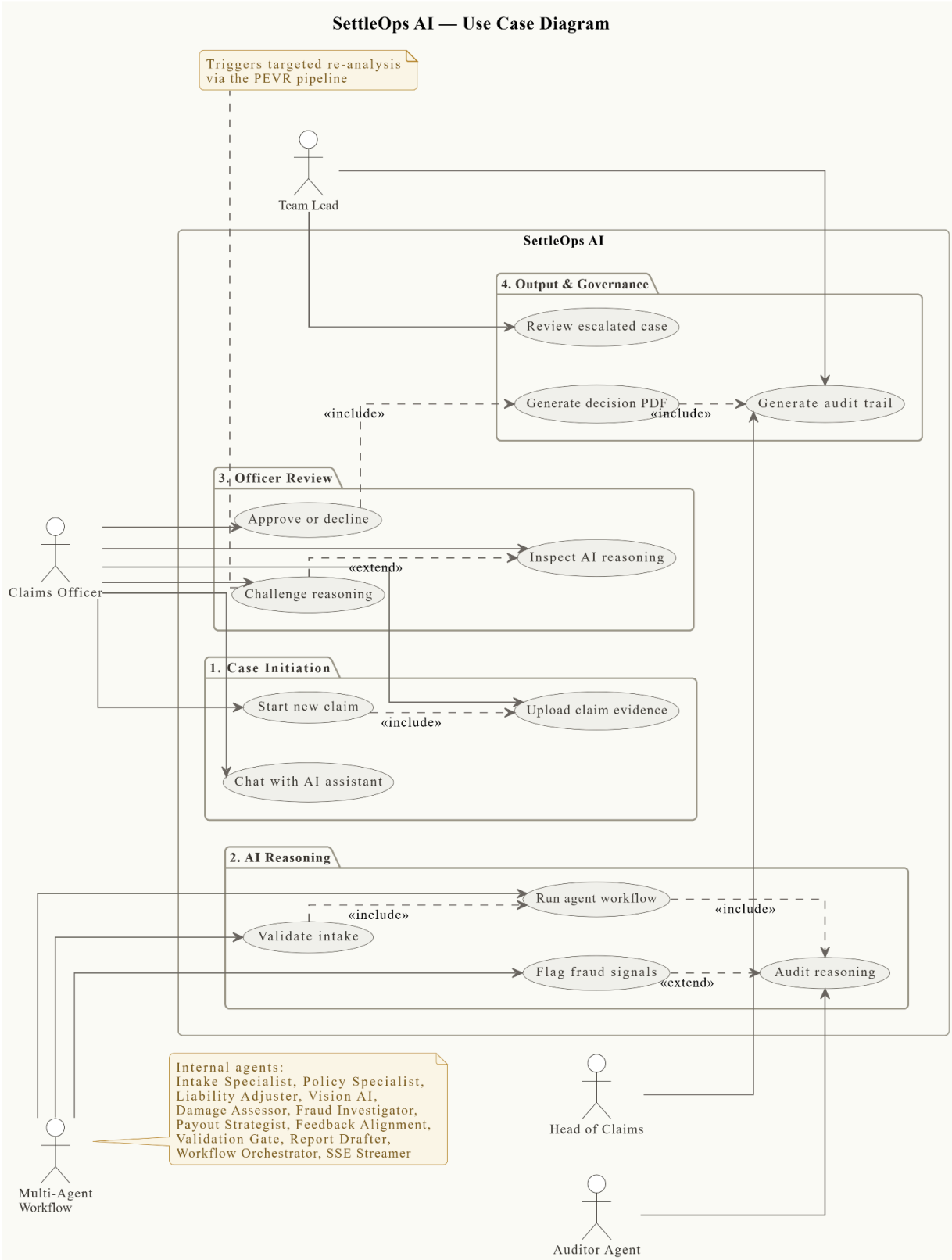

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6. Features Included (Scope Definition)

- A multi-agent AI automation workflow (PEVR*Plan, Execute, Verify, Replan pipeline) that processes claims through stages such as parsing evidence, evaluating key factors (coverage, liability, fraud, payout), validating via an Auditor Agent, and generating a final structured recommendation.
- A three-pane claims officer dashboard that integrates inputs (documents/photos), live AI workflow visualization, and a structured decision blackboard in a single interface.
- A parallel agent reasoning system where multiple specialized agents analyze different aspects of a claim simultaneously to improve speed and accuracy.
- An Auditor Agent validation layer that reviews outputs from all agents, detects inconsistencies, and ensures the final recommendation is coherent and justified.
- A human-in-the-loop feedback mechanism where the officer can identify mistakes or missing reasoning and send feedback to the Auditor Agent.
- A targeted re-analysis workflow, where the Auditor Agent determines which part of the pipeline failed (e.g., liability or fraud) and selectively reruns only the affected components instead of the entire process.
- An AI Strategy Chat interface that allows users to interact with the system for explanations, clarifications, and guided adjustments.
- Automated decision output generation, including Claim Decision PDF and structured audit trail for compliance and review.
- Fraud detection and escalation mechanism that flags suspicious cases and highlights them for further review.

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7. Features Not Included (Scope Control)

- **Full end-to-end insurance system integration (e.g., direct integration with insurer core systems like Merimen eClaims):** Current version focuses on the AI reasoning layer.
- **Automated claim settlement and payment processing:** Final financial execution remains outside system scope.
- **Replacement of licensed loss adjusters:** The system uses adjuster reports as input but does not perform physical inspections or cost estimation independently.
- **Fully autonomous decision-making without human approval:** All decisions require claims officer validation.
- **Advanced predictive fraud network across multiple insurers:** Cross-company intelligence is planned for future phases.
- **Multi-user real-time collaboration on the same case:** Initial version assumes single-user interaction per case.
- **Production-grade security, authentication, and role-based access control:** Simplified for hackathon/demo purposes.
- **Support for non-motor insurance domains:** Current scope is focused on Malaysian motor insurance claims only.

8. Assumptions & Constraints

- **LLM Cost Constraint:** The system assumes moderate-to-high token usage per claim due to multi-document processing and multi-agent reasoning. To manage cost, it uses selective re-execution in the PEVR pipeline (only rerunning affected components),

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document chunking and summarization, and caching of intermediate structured outputs to avoid redundant processing.

- **Technical Constraints:** The current system focuses on the AI reasoning layer and does not fully integrate with production insurance systems. Vision capabilities support analysis but do not replace professional assessment, and features like digital signatures are conceptual for real-world deployment but not fully implemented in the demo.
- **Performance Constraints:** The system targets decision generation within 90 seconds, but performance may vary depending on document size and number of re-analysis cycles. Complex cases and multiple agent interactions may introduce additional latency.
- **User Input:** The system depends on user-provided claim data, and incomplete or low-quality inputs can affect accuracy. Human oversight is required, as final decisions must be approved by claims officers rather than fully automated.

9. Risks & Questions Throughout Development

- **Reasoning Consistency Risk:** There is a risk that different AI agents may produce conflicting outputs, requiring strong coordination and validation mechanisms to maintain consistency across the workflow.
- **Auditor Effectiveness Risk:** If the Auditor Agent fails to detect errors or inconsistencies, incorrect recommendations may pass through, raising concerns about reliability in complex cases.
- **Over-Reliance on AI Risk:** Users may become too dependent on AI-generated decisions, potentially reducing critical thinking and manual verification over time.

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- **Input Quality Risk:** Poor, incomplete, or conflicting claim data can lead to inaccurate analysis, making input quality a key dependency for system performance.
- **Vision Interpretation Risk:** The Vision AI Agent may misinterpret images or fail to confidently assess damage patterns, which could affect downstream reasoning.
- **Workflow Stability Risk (PEVR Pipeline):** When rerunning parts of the pipeline, there is a risk that the system incorrectly identifies the failing component, leading to inefficient or inaccurate re-analysis.
- **Scalability Risk:** Handling many claims simultaneously may increase system load, affecting response time and cost efficiency if not properly optimized.
- **Regulatory & Compliance Risk:** AI-generated decisions must meet legal and regulatory standards, and any lack of compliance could limit real-world adoption.

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